Comparative Robustness and Generalization Analysis of Standard and Attention U-Net Architectures for Polyp Segmentation Under Endoscopic Distribution Shifts

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Repository: https://github.com/313danish/PhD_assignment.git

***Abstract*—The automated segmentation of the colon's polyps in colonoscopy images is a key component for computer aided diagnosis and early intervention for colorectal carcinoma. Current encoder-decoder models, like U-Net, have shown competitive accuracy, but their performance in the presence of the degradations under realistic endoscopy conditions is still not well studied. In this paper, we perform a real-sample comparison between the conventional U-Net and Attention U-Net under Kvasir-SEG. Evaluated on 5 performance metrics: Dice Similarity Coefficient, Intersection-over-Union (IoU), Precision, Recall, and 95th percentile Hausdorff Distance (HD95) are used. Moreover, we systematically test both architectures using five synthetic operational corruptions: additive Gaussian sensor noise (5% variance and 10% variance), Gaussian blur, contrast attenuation, and intensity normalization shifts. Our results showed that Baseline U-Net achieved better boundary precision and anti-high-frequency noise performance (0.7321 Dice at 10% noise vs. 0.6659 Dice) while Attention U-Net achieved better structural robustness against spatial blur (0.6993 Dice vs. 0.6818 Dice) and anti-illumination shift performance (0.7350 Dice vs. 0.7231 Dice) with higher clinical recall.**

***Index Terms*—Biomedical Image Segmentation, U-Net, Attention Mechanism, Endoscopy, Kvasir-SEG, Model Robustness.**

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I. Introduction

Colorectal cancer (CRC) ranks among the primary drivers of oncological mortality worldwide, representing the third most diagnosed malignancy and the second leading cause of cancer-related fatalities. The clinical progression of CRC predominantly initiates from benign adenomatous polyps traversing the adenoma-carcinoma sequence over a multi-year latency period. It is still the clinical "gold standard" to perform regular endoscopy and aggressive mucosal resections to decrease the incidence of the disease and patient mortality. Although conventional colonoscopy is an effective diagnostic tool, there is still significant inter-operator variation, and the rate of adenoma miss is reported to be between 20% and 26%. These diagnostic failures are often due to visual fatigue, high speed of endoscope withdrawal, sessile or flat polyp shape, and motion-related optical artefacts, such as motion blur, fluid puddles and irregular illumination of the lumen. Deep convolutional neural net-works (CNNs) can be used in computer-Aided Detection and Segmentation (CADx) systems to reduce the number of diagnostic errors. Semantic segmentation delivers dense information on the boundaries of lesions, at the level of individual pixels, which can be used as an important quantitative guide for assessing the margins of polyps before electrosurgical resection. The contracting-expanding paradigm combined with skip connections gained momentum in the foundations of the U-Net architecture and became a solid starting point for most biomed-ical segmentation tasks. However, conventional long-range skip connections make direct concatenation between early spatial features and deep semantic representations. This injects non-pathological variations of the mucosa and specular highlights and illumination noise in the large decoder of endoscopic imaging. In order to tackle this problem, soft spatial attention gates are added to the skip pathways in Attention U-Net. Attention gates learn to ignore feature activations in irrelevant background areas, and to accentuate salient gradients around lesion boundaries by leveraging higher-level, coarser gating signals from deeper network layers. Real-world endoscopic deployment is subject to continuous optical and physical distribution changes, that are not captured in standard benchmarks, which report validation metrics on test splits that have been cleaned and curated. Typical problems in endoscopes include lens smudging, different camera to mucosa working distance, focused poor lighting in the endoscopes and sensor thermal noise. Therefore, algorithms with high Dice overlap when tested on clean benchmarks can perform very poorly on the real endoscopic set-ups. This gap is addressed in this paper, which offers an empirical comparative assessment of standard U-Net with Attention U-Net, trained from scratch on Kvasir-SEG benchmark. Both architectures are tested on five synthetic clinical corruptions: additive Gaussian sensor noise (5% and 10% variances), Gaussian motion blur, contrast attenuation, and non-linear illumination shifts, with systematic testing. We were able to show the trade-offs between linear concatenation and non-linear feature gating in realistic domain

perturbations.

II. Methodology

A. Architectural Implementations

Both networks are implemented with identical topological depth, comprising four downsampling stages, a bottleneck latent block, and four upsampling stages. Channel dimensions scale as $[64, 128, 256, 512, 1024]$.

1) Standard U-Net: Each encoder block $l \in \{1, 2, 3, 4\}$ applies two consecutive 3×3 convolutions, each accompanied by 2D Batch Normalization and Rectified Linear Unit (ReLU) activations:

$$\mathbf{z}^l = \text{ReLU}(\text{BN}(\mathbf{W}_2^l * \text{ReLU}(\text{BN}(\mathbf{W}_1^l * \mathbf{x}^l)))) \tag{1}$$

Spatial downsampling between stages is executed via 2×2 max-pooling with stride 2. In the expansive decoder, spatial dimensions are reconstructed using bilinear interpolation followed by a 1×1 convolution, halving the channel capacity. The upsampled feature map is directly concatenated with the corresponding horizontal encoder feature map $\mathbf{x}^l$ via standard skip connections:

$$\mathbf{y}^l = [\mathcal{U}(\mathbf{z}^{l+1}); \mathbf{x}^l] \tag{2}$$

Consider where $\mathcal{U}(\cdot)$ to be the spatial upsampling operator and $[\cdot;\cdot]$ to be the up/down sampling mask element channel-wise concatenation.

2) Attention U-Net: Rather than transferring unfiltered spatial features, Attention U-Net filters skip features $\mathbf{x}^l$ using an Additive Attention Gate (AG). The gate accepts two distinct inputs: the low-level encoder feature map $\mathbf{x}^l \in \mathbb{R}^{F_l \times H \times W}$ and a gating vector $\mathbf{g} \in \mathbb{R}^{F_g \times H \times W}$ derived from the deeper decoder stage which consists of coarser but semantically rich contextual cues.

The spatial alignment and attention weight matrix $\boldsymbol{\alpha}^l \in [0, 1]^{1 \times H \times W}$ are computed as:

$$\mathbf{q}_{\text{att}}^l = \text{ReLU}\left(\mathbf{W}_x^T \mathbf{x}^l + \mathbf{W}_g^T \mathbf{g} + \mathbf{b}_g\right) \tag{3}$$

$$\boldsymbol{\alpha}^l = \sigma\left(\boldsymbol{\psi}^T \mathbf{q}_{\text{att}}^l + b_\psi\right) \tag{4}$$

where $\mathbf{W}_x \in \mathbb{R}^{F_l \times F_{\text{int}}}$, $\mathbf{W}_g \in \mathbb{R}^{F_g \times F_{\text{int}}}$, and $\boldsymbol{\psi} \in \mathbb{R}^{F_{\text{int}} \times 1}$ are implemented via 1×1 2D convolutions, $\sigma(t) = \frac{1}{1+e^{-t}}$ is the sigmoid activation function, and F_{int} denotes the intermediate projection subspace (configured to $F_l/2$). The filtered feature map $\hat{\mathbf{x}}^l$ passed to the decoder is the element-wise Hadamard product:

$$\hat{\mathbf{x}}^l = \boldsymbol{\alpha}^l \odot \mathbf{x}^l \tag{5}$$

B. Loss Function Formulation

Due to the high foreground-background class imbalance where polyps typically occupy less than 15% of the total pixel volume in a colonoscopy frame, utilizing standard Binary Cross-Entropy (BCE) in isolation produces model bias toward the non-lesion background class. We formulate a composite objective function combining BCE and soft Dice Loss:

$$\mathcal{L}_{\text{total}}(\mathbf{y}, \hat{\mathbf{y}}) = \mathcal{L}_{\text{BCE}}(\mathbf{y}, \hat{\mathbf{y}}) + \mathcal{L}_{\text{Dice}}(\mathbf{y}, \hat{\mathbf{y}}) \tag{6}$$

The binary cross-entropy loss provides uniform pixel-level gradient stability:

$$\mathcal{L}_{\text{BCE}}(\mathbf{y}, \hat{\mathbf{y}}) = -\frac{1}{N} \sum_{i=1}^{N} [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \tag{7}$$

while the soft Dice loss optimizes global spatial overlap directly:

$$\mathcal{L}_{\text{Dice}}(\mathbf{y}, \hat{\mathbf{y}}) = 1 - \frac{2 \sum_{i=1}^{N} y_i \hat{y}_i + \epsilon}{\sum_{i=1}^{N} y_i + \sum_{i=1}^{N} \hat{y}_i + \epsilon} \tag{8}$$

where $y_i \in \{0, 1\}$ represents the ground truth label at pixel i, $\hat{y}_i = \sigma(z_i) \in [0, 1]$ represents the predicted sigmoid activation probability, and $\epsilon = 10^{-7}$ prevents division by zero.

C. Experimental Setup

The benchmark is evaluated on the Kvasir-SEG dataset, containing 1,000 clinically validated gastrointestinal images. A deterministic 80/20 train-validation split is enforced (800 training images, 200 validation images).

- **Pre-processing:** All images and masks are resized to 256×256 pixels and normalized using ImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$).
- **Training and Augmentations:** The networks are both trained from scratch with the Adam optimizer ($\eta = 1 \times 10^{-4}$) with a batch size of 4 for 15 epochs on an NVIDIA GPU platform, and both are trained with the same augmentation settings, which include random horizontal flips ($p = 0.5$), random vertical flips ($p = 0.5$), and random 90-degree rotations ($p = 0.5$).
- **Optimization:** Both networks are trained from scratch with the Adam optimizer ($\eta = 1 \times 10^{-4}$), and batch size of 4, for 15 epochs on an NVIDIA GPU platform.

III. Experimental Results

A. Clean Benchmark Evaluation

Table I outlines the evaluation metrics across the uncorrupted validation partition.

TABLE I
Quantitative Performance on Clean Kvasir-SEG Validation Data

Architecture	Dice	IoU	Precision	Recall	HD95 (px)
Standard U-Net	**0.7439**	**0.6134**	**0.8052**	0.7231	**31.76**
Attention U-Net	0.7388	0.6034	0.7799	**0.7350**	34.58

Standard U-Net achieves a marginally higher Dice score (0.7439 vs. 0.7388), IoU (0.6134 vs. 0.6034), and boundary alignment with an HD95 of 31.76 pixels versus 34.58 pixels. Conversely, Attention U-Net demonstrates superior sensitivity, achieving a Recall of 0.7350 compared to 0.7231 for the baseline. In medical diagnostic frameworks, sensitivity is clinically paramount, as failing to segment subtle or flat adenomas (false negatives) carries severe prognostic consequences compared to modest over-segmentation (false positives).

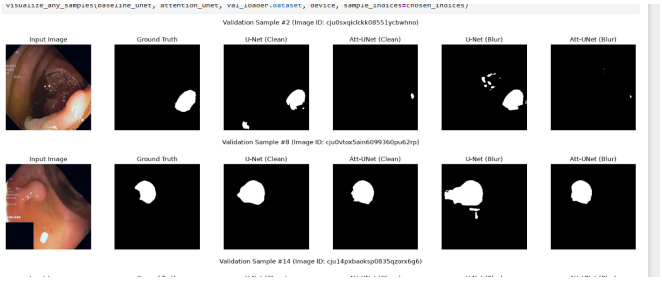


Fig. 1. Qualitative comparative segmentation results on Kvasir-SEG validation samples under clean and blurred operational conditions.

B. Robustness and Corruption Profiling

To measure real-world reliability, both trained models were evaluated under five distinct visual corruptions without re-training:

1) Additive Gaussian Noise (5% variance)
2) Additive Gaussian Noise (10% variance)
3) Gaussian Blur (5×5 kernel, $\sigma = 1.0$)
4) Contrast Attenuation (40% dynamic range reduction)
5) Non-linear Intensity Shift (additive offset $+0.5$, scaled by 0.8)

TABLE II
ROBUSTNESS EVALUATION ACROSS SYNTHETIC CORRUPTIONS

Test Condition	**Standard U-Net**		**Attention U-Net**	
	Dice	**IoU**	**Dice**	**IoU**
Clean Baseline	0.7439	0.6134	0.7388	0.6034
Gaussian Noise (5%)	0.7450	0.6150	0.7287	0.5897
Gaussian Noise (10%)	**0.7321**	**0.5986**	0.6659	0.5123
Gaussian Blur (5×5)	0.6818	0.5320	**0.6993**	**0.5551**
Contrast Attenuation ($\times 0.6$)	0.6717	0.5214	**0.6859**	**0.5400**
Intensity Norm Shift	0.7016	0.5609	**0.7079**	**0.5649**

The degradation profiles in Table II show strong differences between the two behavioral characteristics:

- **Low-Frequency Invariance (Blur, Contrast, Illumination):** Attention U-Net outperforms standard U-Net consistently on optical degradations in terms of Dice and IoU, achieving Dice scores of 0.6993 vs. 0.6818 for Gaussian Blur; 0.6859 vs. 0.6717 for Contrast Attenuation; and 0.7079 vs. 0.7016 for Intensity Shift. The attention gating mechanism can effectively focus on the body of a lesion even with low resolution edge gradients from the coarse semantic vectors extracted from the latent bottlenecks.
- **High-Frequency Stochastic Noise:** By contrast, the performance of standard U-Net is significantly better with additive noise. At 10% Gaussian noise, standard U-Net preserves a Dice of 0.7321 (1.58% relative decrease), while Attention U-Net has a massive dropout ratio of 0.6659 (9.87% relative decrease). The linear 1×1 projections within the attention gates amplify uncorrelated pixel-level perturbations, disrupting the calculated gating coefficients.

IV. DISCUSSION AND CLINICAL IMPLICATIONS

The obtained experimental results show that the attention gating isn't an unconstrained enhancement on all operational criteria. While standard U-Net has a better Inductive Bias against high-frequency sensor noise, Attention U-Net provides critical clinical utility:

- **Diagnostic Sensitivity:** The U-Net is sensitive, with a Diagnostic Sensitivity of: The higher the recall on clean data (0.7350), the more re it showed. The more re it showed, the higher the recall on clean data (0.7350), duced rates of missed lesions.
- **Endoscopic Environment Match:** The procedure of real-world pro The possibility of using cesure and unclear images due to motion blur and low illuminance tion from fluid pools or distance variations occur far more frequently than pure stochastic sensor noise. Attention U-Net is more stable in blur and low light environments. The resulting contrast conditions is better for colonoscopy video streams.

V. CONCLUSION

This study proposed a comparative standard U-Net and Attention U-Net benchmark and corrup- tion stress test on the Kvasir-SEG dataset. Standard U-Net shows superior robustness to additive Gaussian noise while Attention U-Net shows superior performance in resisting the optical blurring. Other factors, such as contrast attenuation and shifts in illumination must be considered rior clinical recall. Future studies will examine the use of additive noise stable but highly boundary sensitive attention paths by adding noise-resilient gating mechanisms to such attention paths (spectral pooling), and spatial dropout.